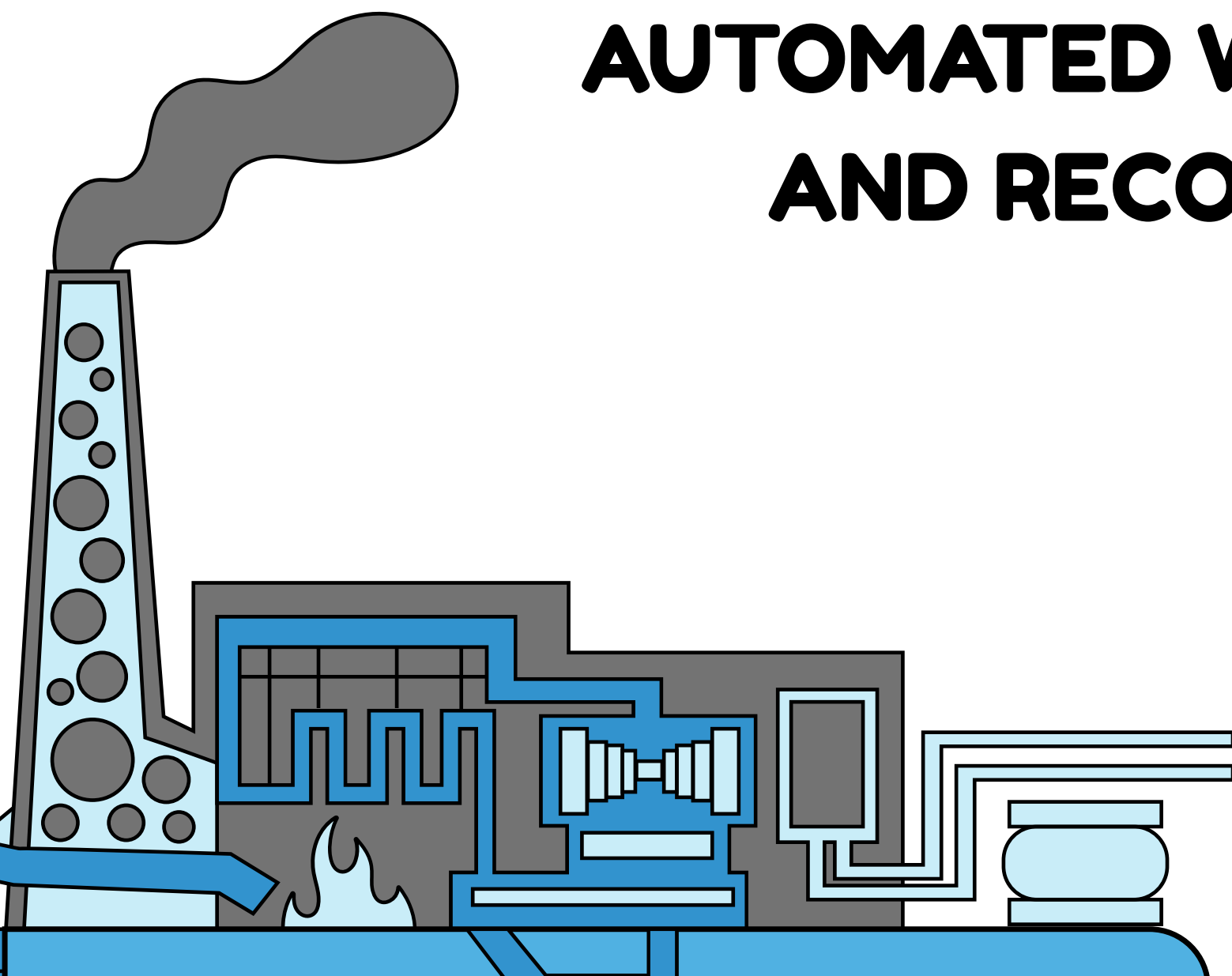
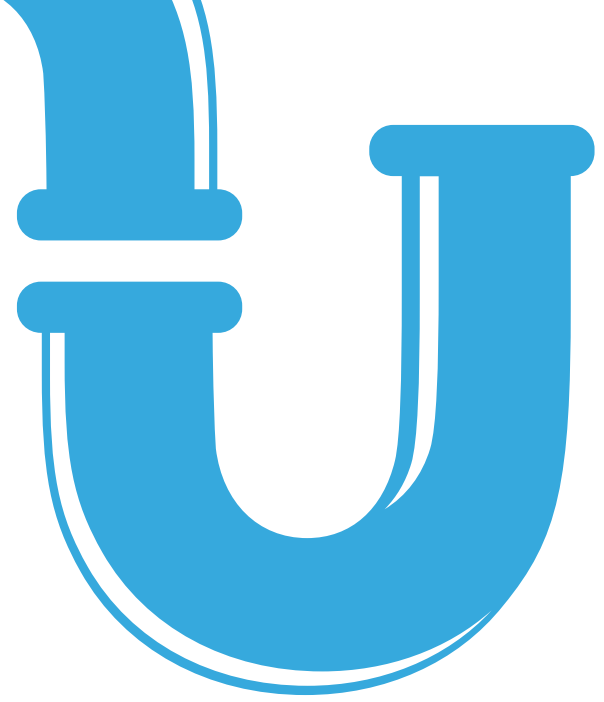




AUTOMATED WATER FILTRATION DESIGN AND RECOMMENDATION SYSTEM





Team Members:

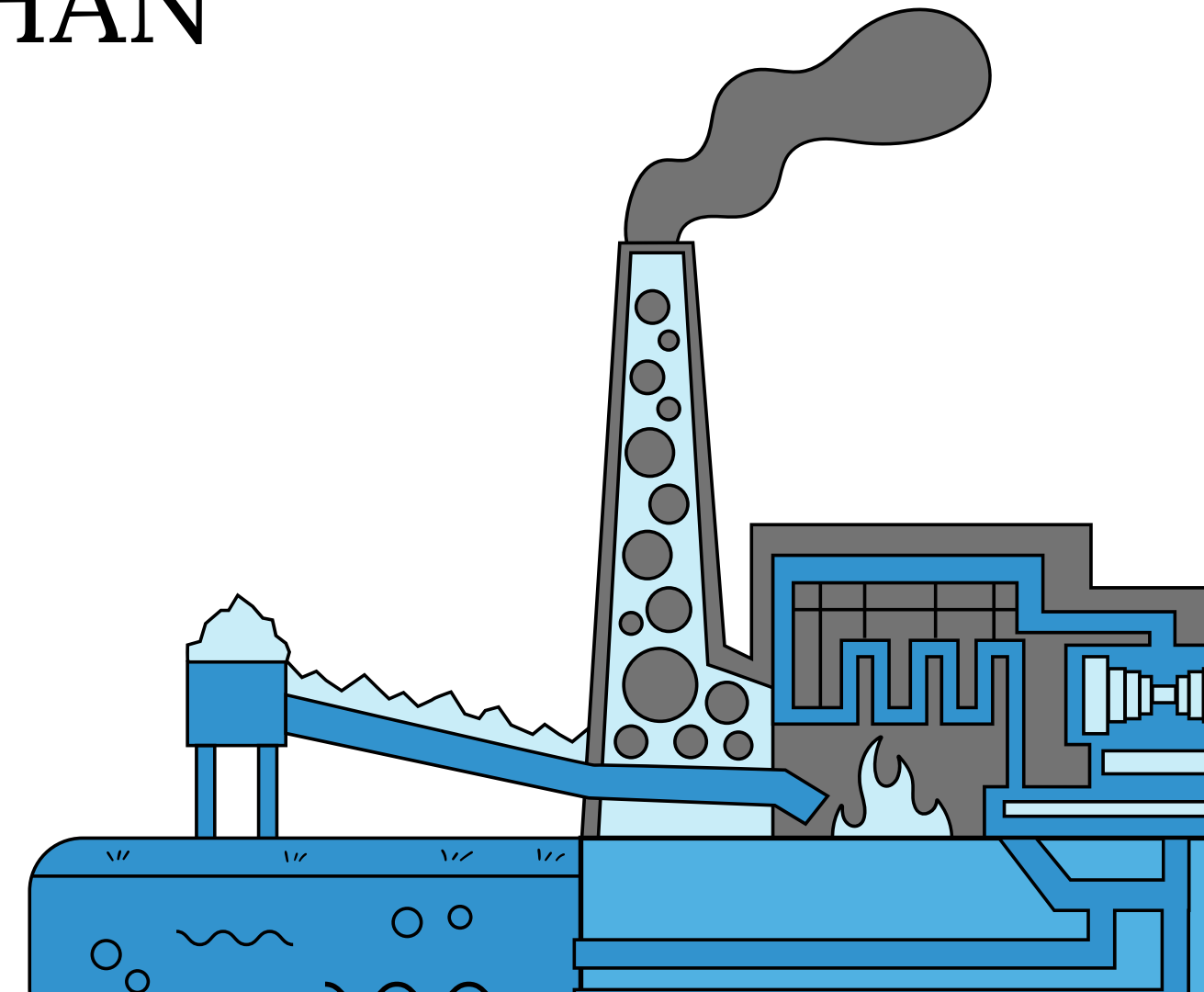
ABHIRAMI R

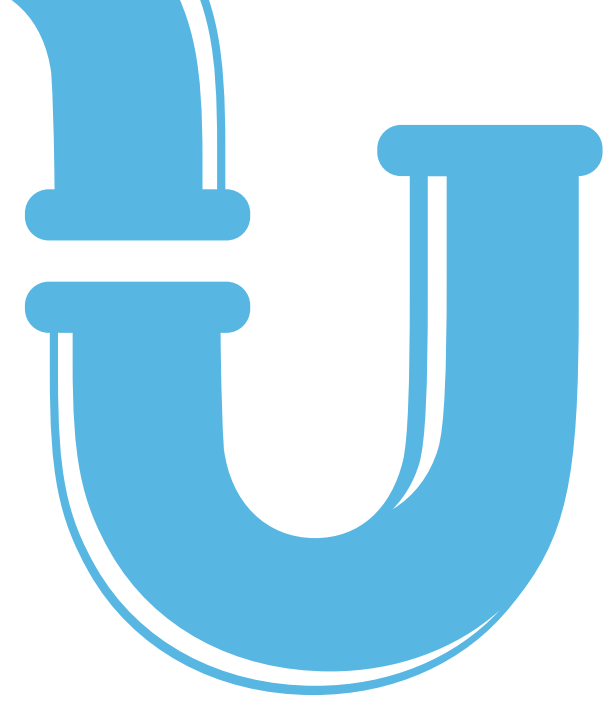
ALAKANANADA J NAIR

GAYATHRI K GOPINATHAN

AJIT GURU

ARJUN GANESHAN



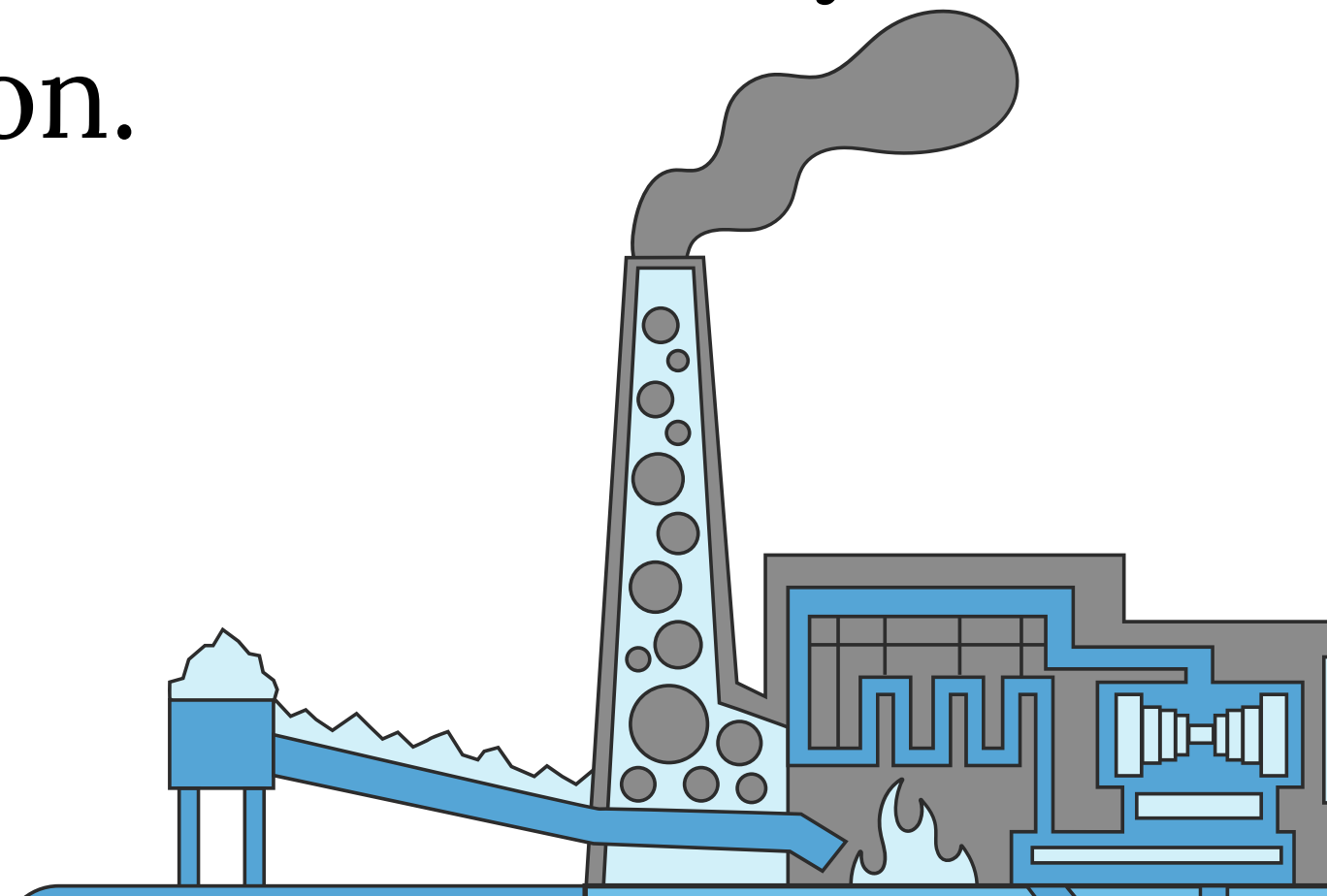


Introduction



RELEVANCE OF THE PROJECT

Access to clean and safe drinking water is a basic necessity for human health. However, water quality varies greatly from place to place due to factors such as pollution, hardness, salinity, and microbial contamination.





Many people are unaware of the exact quality of water they use and often choose water filters without proper knowledge, which may not effectively solve their water problems.

This project is relevant because it helps users identify suitable water filtration solutions based on specific water quality parameters. By automating the analysis and recommendation process, the system ensures better decision-making, cost efficiency, and improved access to safe drinking water.



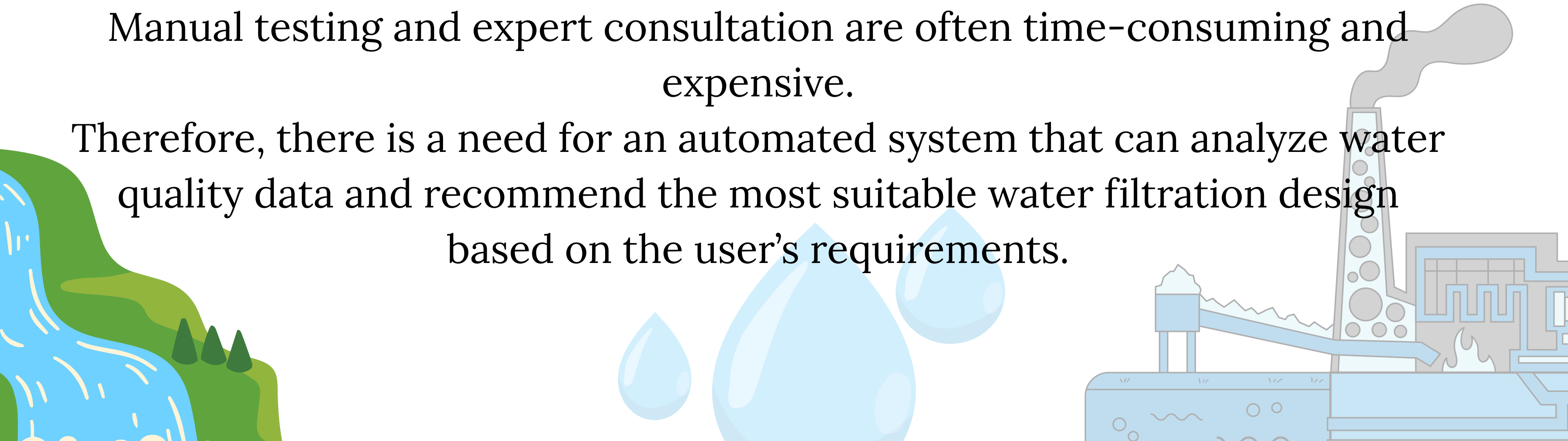


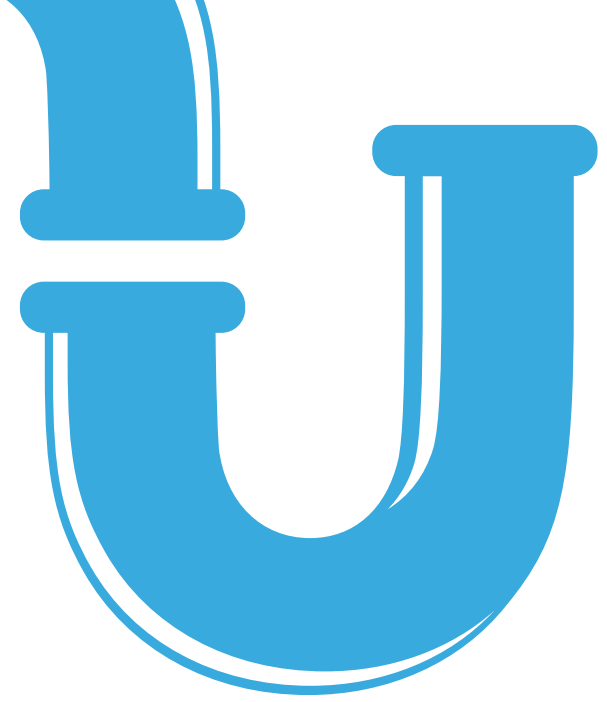
PROBLEM STATEMENT

Selecting an appropriate water filtration system is a challenging task for users due to a lack of technical knowledge about water quality parameters and filtration technologies. Most people rely on generic recommendations or marketing information, which may not address their actual water issues.

Manual testing and expert consultation are often time-consuming and expensive.

Therefore, there is a need for an automated system that can analyze water quality data and recommend the most suitable water filtration design based on the user's requirements.





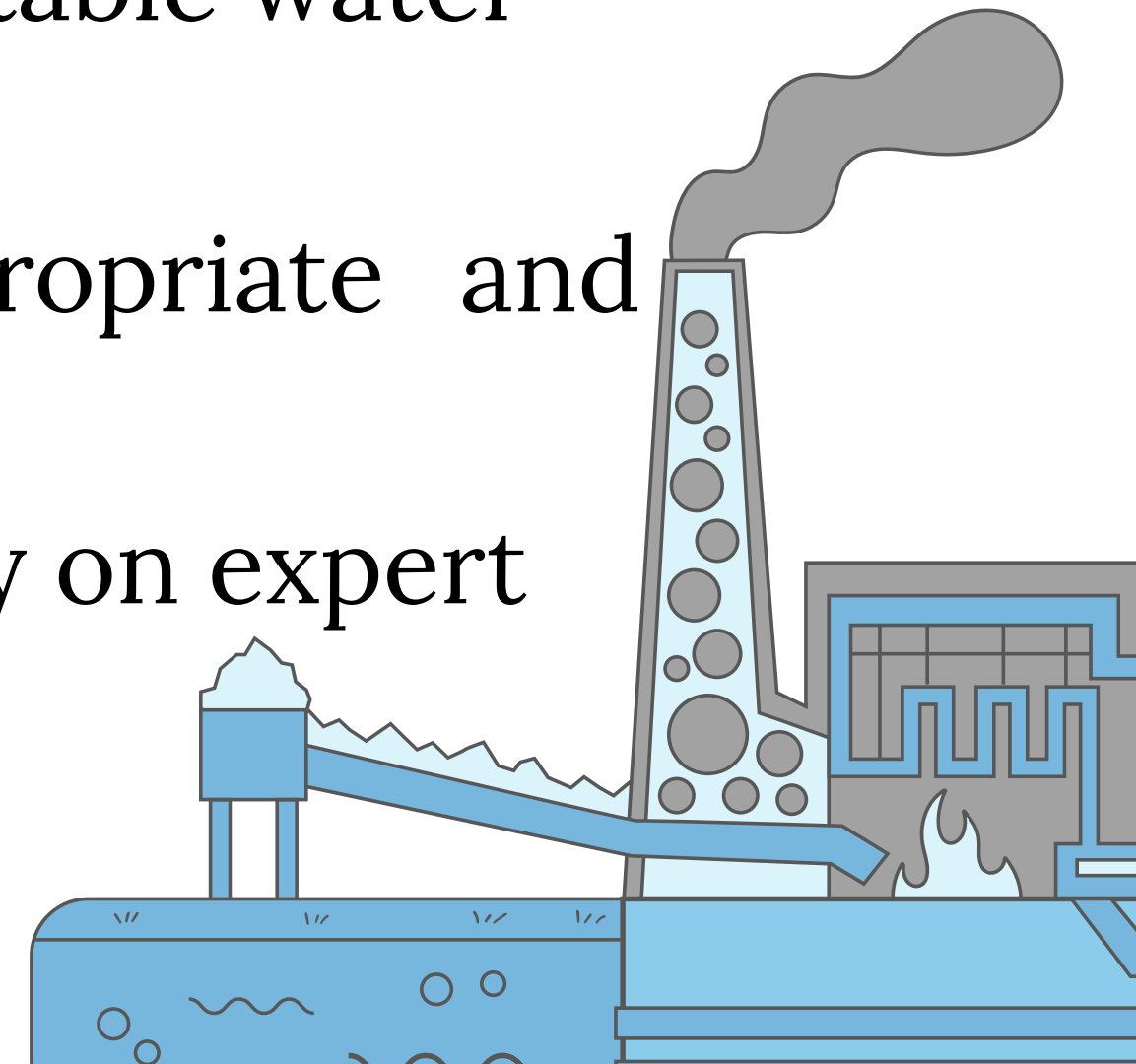
OBJECTIVES AND SCOPE

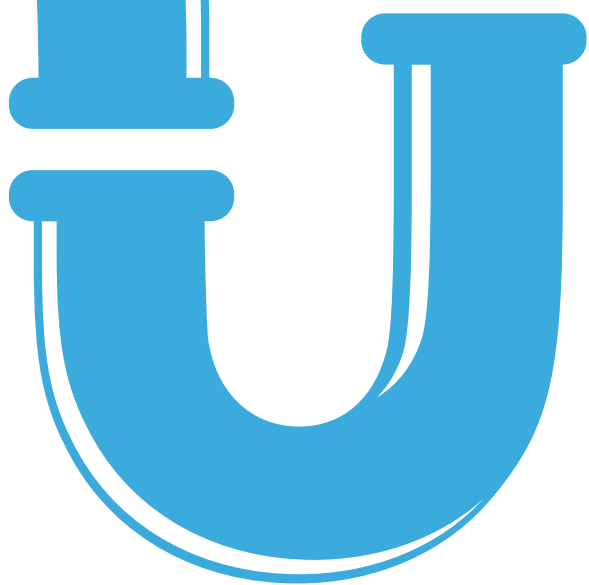
Objective



The main objectives of this project are:

- To analyze water quality parameters effectively.
- To automate the recommendation of suitable water filtration systems.
- To help users choose the most appropriate and efficient filtration solution.
- To reduce manual effort and dependency on expert consultation.

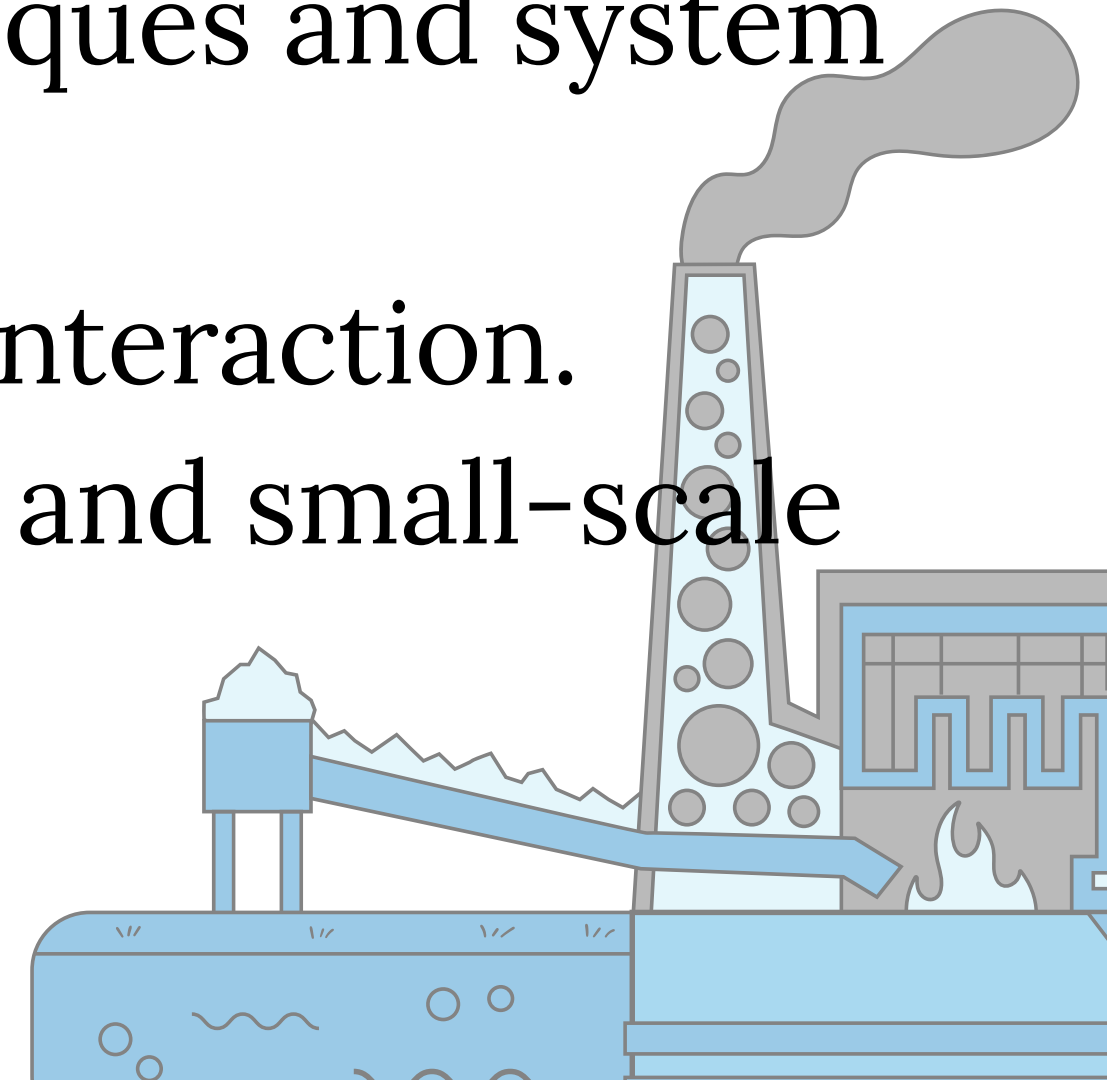




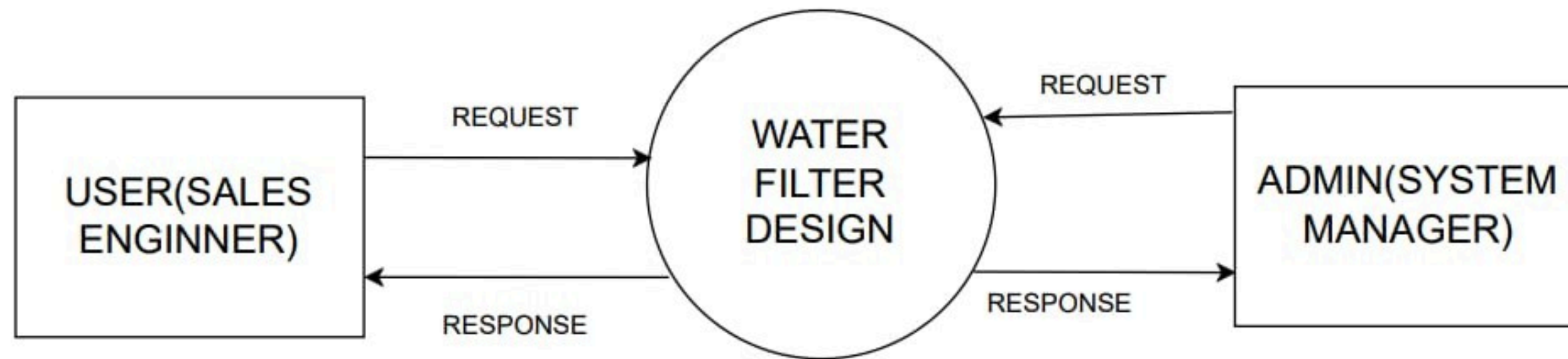
Scope

The scope of this project includes:

- Accepting water quality inputs such as pH, hardness, turbidity, and contamination levels.
- Recommending appropriate filtration techniques and system designs.
- Providing a user-friendly interface for easy interaction.
- Supporting decision-making for households and small-scale applications.

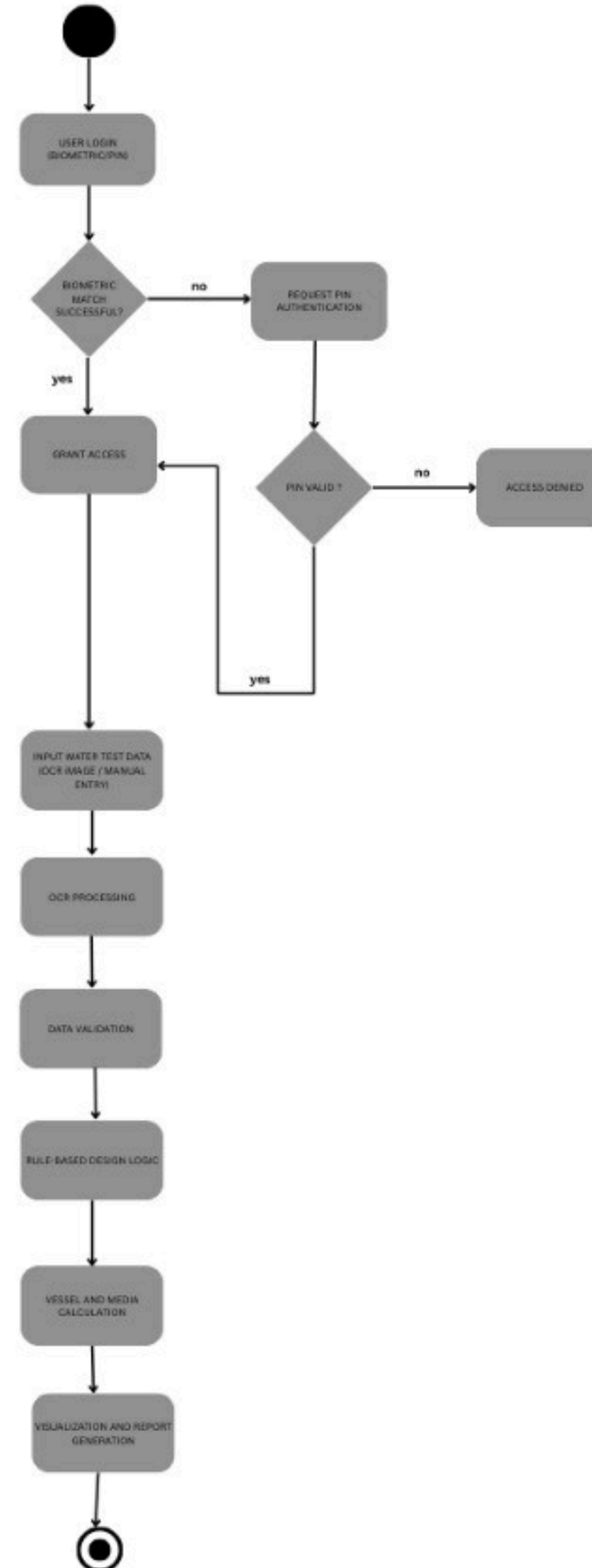


DFD LEVEL 0

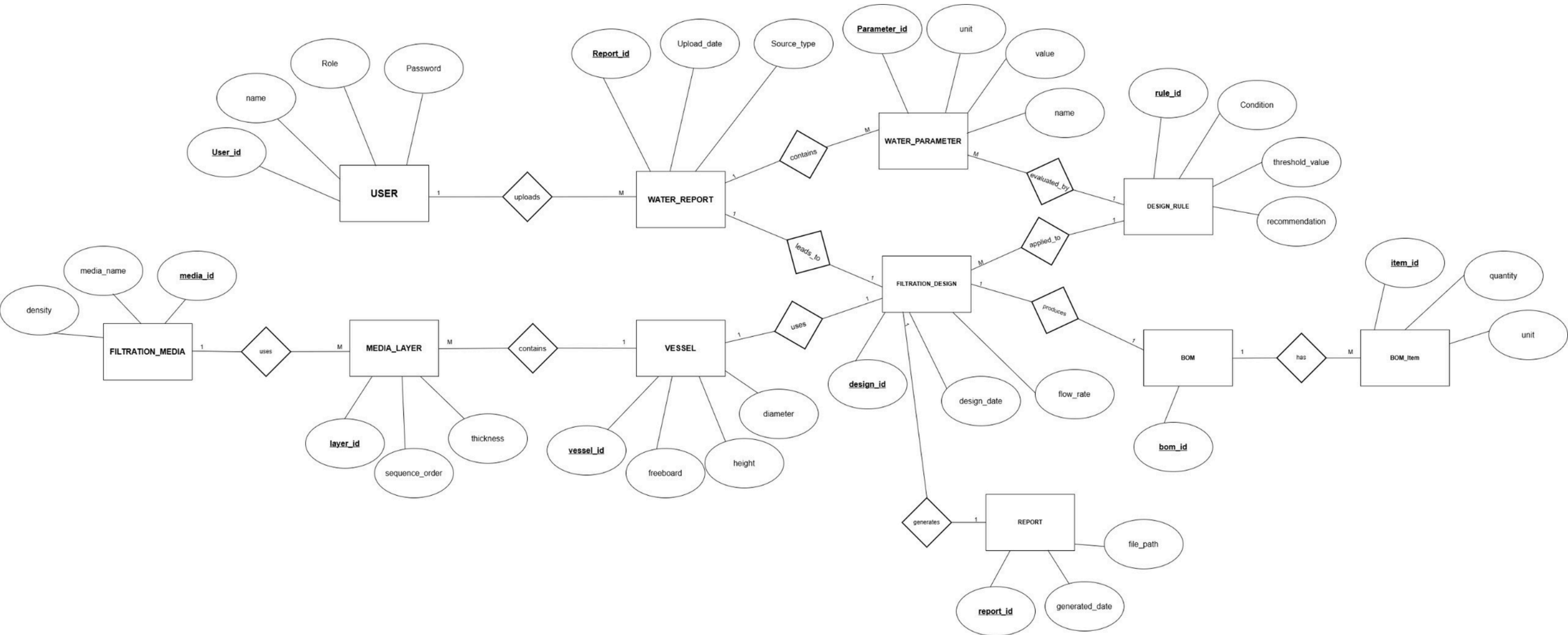


DFD LEVEL1

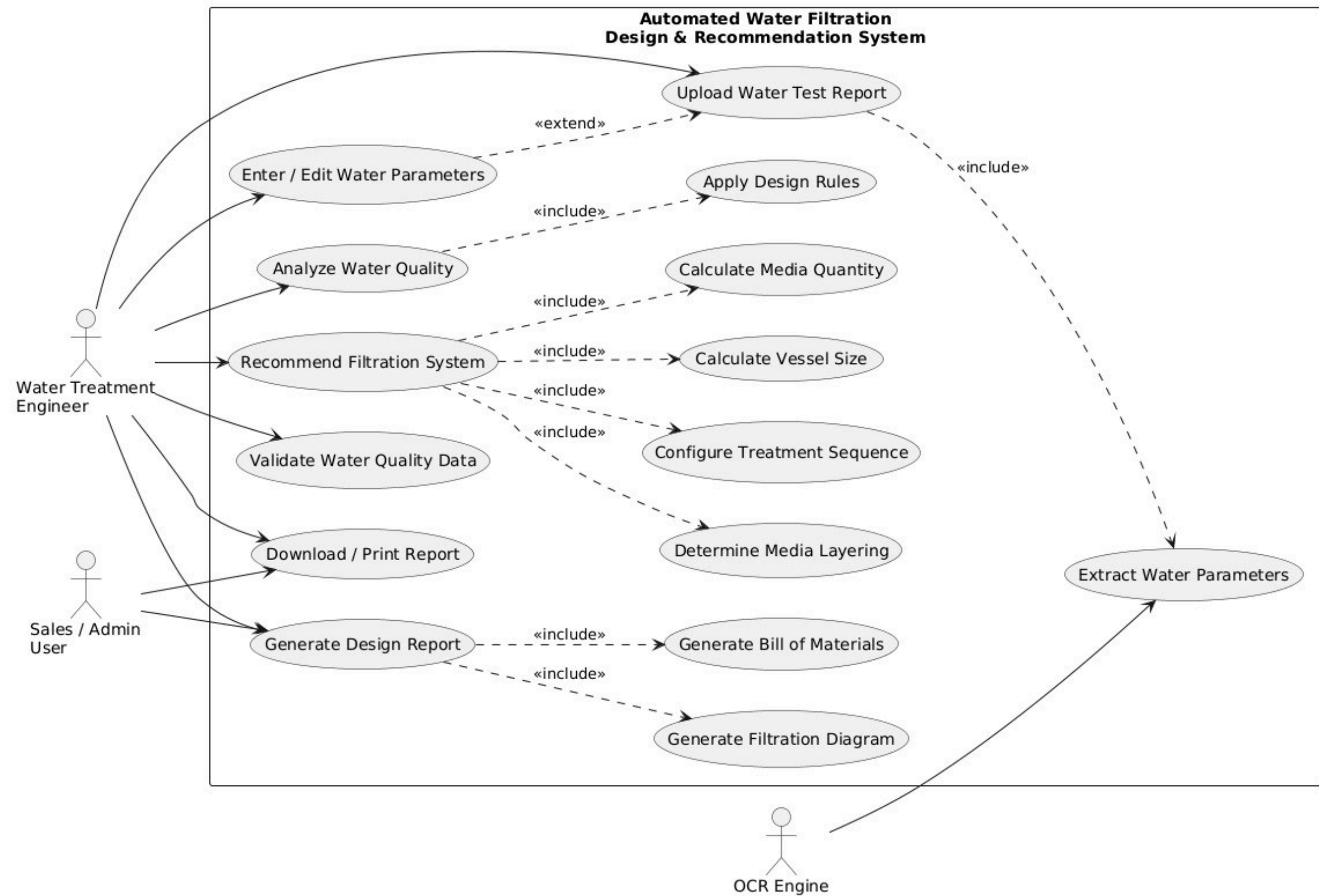
WORK FLOW DIAGRAM



ERD DIAGRAM



USE CASE DIAGRAM





THANK YOU

